

$$\begin{aligned}\therefore x + (x + 8) + (x + 15) &= 53 \\ \Rightarrow x &= 10 \\ \therefore A : B : C &= (10 + 15) : (10 + 8) : 10 \\ &= 25 : 18 : 10\end{aligned}$$

43. (b) Let Q join for x months.

$$\begin{aligned}\therefore \text{Ratio of capital} &= 2525 \times 12 : 1200 \times x \\ &= 2525 : 100x = 101 : 4x\end{aligned}$$

$$\therefore P\text{'s profit} = \frac{101}{101 + 4x} \times 1644$$

$$\Rightarrow 1212 = \frac{101 \times 1644}{101 + 4x}$$

$$\Rightarrow \frac{1}{137} = \frac{1}{101 + 4x}$$

$$\Rightarrow 4x = 36 \Rightarrow x = 9$$

Hence, Q joined for 9 months i.e. he joined after 3 months.

44. (a) Originally, let the numbers of seats for Mathematics, Physics and Biology be $5x$, $7x$ and $8x$, respectively.

Number of increased seats are (140% of $5x$) (150% of $7x$) and (175% of $8x$).

Required ratio can be obtained as

$$\begin{aligned}\Rightarrow \left(\frac{140}{100} \times 5x\right) : \left(\frac{150}{100} \times 7x\right) : \left(\frac{175}{100} \times 8x\right) \\ \Rightarrow \left(7x : \frac{21}{2}x : 14x\right) \times 2 \\ \Rightarrow (14x : 21x : 28x) \div 7x \\ \therefore 2 : 3 : 4\end{aligned}$$

45. (d) Let the number of passengers travelling by class I and class II be x and $50x$ respectively.

Then, amount collected from class I and II will be ₹ $3 \times x$ and ₹ $50x$ respectively.

$$\text{Given, } 3x + 50x = 1325$$

$$53x = 1325 \Rightarrow x = 25$$

$\therefore$ Amount collected from class II

$$= 50 \times 25 = ₹ 1250$$

46. (b) As $x \propto \frac{1}{y^2} \Rightarrow x = \frac{k}{y^2}$... (i)

$$\text{If } x = 1$$

$$\text{and } y = 6$$

$$1 = \frac{k}{6^2} \Rightarrow k = 36$$

On putting the value of k in Eq. (i), we get

$$x = \frac{36}{y^2} \quad \dots (ii)$$

$$\text{I. On putting } y = 3 \text{ in Eq. (ii), } x = \frac{36}{9}$$

$$x = 4$$

II. On putting $y = 6$ in Eq. (ii), we get

$$x = \frac{36}{36} = 1$$

$$x = 1$$

Both statements I and II are correct.

47. (d) Given that, $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = \frac{1}{2}$

$$\Rightarrow a = \frac{b}{2}, c = \frac{d}{2}, e = \frac{f}{2}$$

$$\begin{aligned}\text{I. } \frac{3a + 5c + 7e}{3b + 5d + 7f} &= \left(\frac{\frac{3b}{2} + \frac{5d}{2} + \frac{7f}{2}}{3b + 5d + 7f} \right) \\ &= \frac{1}{2} \frac{(3b + 5d + 7f)}{3b + 5d + 7f} = \frac{1}{2}\end{aligned}$$

$$\text{II. } \frac{a^2 + c^2 + e^2}{b^2 + d^2 + f^2} = \frac{\frac{b^2}{4} + \frac{d^2}{4} + \frac{f^2}{4}}{b^2 + d^2 + f^2} = \frac{1}{4}$$

$\therefore$ Neither I nor II correct.

48. (c) I. 4 leaps of cat = 3 leaps of dog

$$\Rightarrow 1 \text{ leap of cat} = \frac{3}{4} \text{ leap of dog}$$

Cat takes 5 leap for every 4 leaps of dog.

$\therefore$ Required Ratio

$$= (5 \times \text{cat's leap}) : (4 \times \text{dog's leap})$$

$$= \left(5 \times \frac{3}{4} \text{ dog's leap}\right) : (4 \times \text{dog's leap})$$

$$= 15 : 16$$

$$\text{Thus, } \frac{\text{Speed of cat}}{\text{Speed of dog}} = \frac{15}{16}$$

$$\text{II. } \frac{\text{Distance (cat)}}{\text{Distance (dog)}} = \frac{s(\text{cat}) \times t}{s(\text{dog}) \times t}$$

$$= \frac{s(\text{cat}) \times 30}{s(\text{dog}) \times 30} = \left(\frac{s(\text{cat})}{s(\text{dog})} = \frac{15}{16} \right)$$

Thus, both statements I and II are correct.

49. (a) Number of people having characteristic X

$$= 10 + 30 = 40$$

Number of people having characteristic Y

$$= 10 + 20 = 30$$

$$\text{Required ratio} = 40 : 30 = 4 : 3.$$

50. (c) Fresh grapes contain 10% pulp.

$\therefore$ 20 kg fresh grapes contain 2 kg pulp.

Dry grapes contain 80% pulp.

2 kg pulp would contain

$$\frac{2}{0.8} = \frac{20}{8} = 2.5 \text{ kg dry grapes.}$$

51. (c) Refer to question 12

52. (c) Refer to question 20

$\therefore$ Their product = $18 \times 27 = 486$

53. (b) Let cost price of milk be ₹ x per kg.

Then, according to the rule of mixture,

$$22 : (x - 22) = 15 : 3$$

$$\Rightarrow \frac{22}{x - 22} = \frac{15}{3}$$

$$\Rightarrow \frac{22}{x - 22} = 5$$

$$\therefore x = ₹ 26.40$$

54. (a) Total number of inhabitants = 1935000

Total number of females = 935000

$\therefore$ Total number of males

$$= 1935000 - 935000 = 1000000$$

$$\therefore \text{sex ratio} = \frac{935000}{1000000} \times 1000 = 935$$

55. (d) Given, ratio of boys to girls = 7 : 5

and total number of students = 2400

Sum of ratio = 7 + 5 = 12

$$\therefore \text{Number of girls} = \frac{5}{12} \times 2400 = 1000$$

56. (a) Given, $A : B = 2 : 3$, $B : C = 5 : 7$

and $C : D = 3 : 10$

$$\begin{aligned}\therefore \frac{A}{D} &= \frac{A}{B} \times \frac{B}{C} \times \frac{C}{D} \\ &= \frac{2}{3} \times \frac{5}{7} \times \frac{3}{10} = \frac{1}{7}\end{aligned}$$

$$\therefore A : D = 1 : 7$$

57. (d) Given that, $(x + y) : (x - y) = 3 : 5$

$$\therefore \frac{x + y}{x - y} = \frac{3}{5}$$

On applying componendo and dividendo rule, we get

$$\frac{(x + y) + (x - y)}{(x + y) - (x - y)} = \frac{3 + 5}{3 - 5}$$

$$\Rightarrow \frac{2x}{2y} = \frac{8}{-2} \Rightarrow \frac{x}{y} = -4$$

$$\therefore x = -4y$$

But it is given, xy = positive

$\therefore -4y \times y$ = positive

$\Rightarrow -4y^2$ = positive, which is not possible.

Hence, no real solution for x and y exists.

58. (a) Given, ratio of ages of A and B i.e.

$$A : B = 2 : 5$$

Ratio of ages of B and C i.e.

$$B : C = 3 : 4$$

$$\therefore \text{Ratio of ages of } A, B \text{ and } C = \frac{2}{3} : \frac{5}{4} : 1$$

$$= 2 \times 3 : 3 \times 5 : 5 \times 4 = 6 : 15 : 20$$

59. (a) Let height of tree be h ft and age be a yr.

Then, according to the question,

$$h \propto \sqrt{a} \Rightarrow h = k\sqrt{a} \quad \dots (i)$$

where, k is a constant.

If age = 9 yr, then height = 4 ft

From Eq. (i), we get $4 = k\sqrt{9}$

$$\Rightarrow 4 = k \times 3 \therefore k = \frac{4}{3}$$

If age = 16 yr, then $h = k\sqrt{16} = \frac{4}{3} \times 4$

$$\left[\therefore k = \frac{4}{3} \right]$$

$$\therefore h = \frac{16}{3} \text{ ft} = 5\frac{1}{3} = 5 \text{ ft } 4 \text{ inch}$$

60. (b) Refer to question 34.

61. (c) We have, $\frac{a}{b} = \frac{3}{5}$

$$\therefore b = \frac{5a}{3} \text{ and } \frac{b}{c} = \frac{7}{8}$$

$$\Rightarrow c = \frac{8}{7}b = \frac{8}{7} \times \frac{5a}{3} = \frac{40a}{21}$$

$$\therefore a : b : c = a : \frac{5a}{3} : \frac{40a}{21}$$

$$\text{Now, } 2a : 3b : 7c = 2a : 5a : \frac{40a}{3}$$

$$= 2 : 5 : \frac{40}{3} = 6 : 15 : 40$$

62. (d) Let the distance be x .

Now, ratio of time taken to travel the distance by each bus is

$$\frac{x}{2} : \frac{x}{3} : \frac{x}{4} \Rightarrow \frac{12}{2} : \frac{12}{3} : \frac{12}{4}$$

$$\Rightarrow 6 : 4 : 3$$

63. (c) Since, milk = $7x$ and water = $3x$

$$\text{Then, } 7x + 3x = 30 \Rightarrow 10x = 30$$

$$\therefore x = 3$$

So, milk = 21 L and water = 9 L

Let y L of water to be added.

$$\text{Then, } \frac{21}{9+y} = \frac{1}{2}$$

$$\Rightarrow 9 + y = 42$$

$$\therefore y = 33 \text{ L}$$

64. (a) Refer to question 27.

65. (a) $\frac{a}{b} = \frac{b}{c} = \frac{c}{d} = k$ (say)

$$\Rightarrow a = bk, b = ck, c = dk$$

$$\therefore a = dk^3, b = dk^2, c = dk$$

$$\text{I. } \frac{b^3 + c^3 + d^3}{a^3 + b^3 + c^3}$$

$$= \frac{b^3 + c^3 + d^3}{(bk)^3 + (ck)^3 + (dk)^3}$$

$$= \frac{1(b^3 + c^3 + d^3)}{k^3(b^3 + c^3 + d^3)} = \frac{1}{k^3} = \frac{d}{a}$$

Hence, statement I is correct.

$$\text{II. Similarly } \frac{a^2 + b^2 + c^2}{b^2 + c^2 + d^2} = k^2 = \frac{b}{d}$$

Hence, statement II is incorrect.

13

LOGARITHM

Usually (2-3) questions have been asked from this chapter. Generally questions are based on fundamental rules of logarithm. It is especially used to solve complicated mathematical calculations. Applications of square, cube, surds and indices are extremely used here.



DEFINITION

If 'a' is a positive real number, other than 1 and 'b' is a rational number such that $a^b = N$, then we say that logarithm of N to base 'a' is b or 'b' is the logarithm of N to base 'a', written as $\log_a N = b$

So, $a^b = N \Leftrightarrow \log_a N = b$, e.g. $7^0 = 1 \Leftrightarrow \log_7 1 = 0$, $81^{1/4} = 3 \Leftrightarrow \log_{81} 3 = 1/4$

EXAMPLE 1. Which one of the following has the value of $x = 4$?

a. $\log_2 x = 3$

b. $\log_5 x = 3$

c. $\log_{81} x = \frac{3}{2}$

d. $\log_{\sqrt{2}} x = 4$

Sol. d. a. $\log_2 x = 3 \Leftrightarrow x = 2^3 = 8$ b. $\log_5 x = 3 \Leftrightarrow x = 5^3 = 125$ [by definition]

c. $\log_{81} x = 3/2 \Leftrightarrow x = (81)^{3/2} = (3^4)^{3/2}, \Leftrightarrow x = 3^6 \Leftrightarrow x = 729$

d. $\log_{\sqrt{2}} x = 4 \Leftrightarrow x = (\sqrt{2})^4, x = (2^{1/2})^4 = 2^2 = 4$

EXAMPLE 2. If $\log_5 y = x$ and $\log_2 z = x$, then the value of 20^x in terms of y and z is

a. zy

b. z^2y

c. xy^2

d. z/y

Sol. b. Given, $\log_5 y = x$ and $\log_2 z = x \Rightarrow y = 5^x$ and $z = 2^x$... (i)

Now, $20^x = (2^2 \times 5)^x = 2^{2x} \times 5^x = (2^x)^2 \cdot 5^x = z^2 \cdot y$ [from Eq. (i)]

$\therefore 20^x = z^2 \cdot y$

Fundamental Rules of Logarithms

Rule 1 If m and n are positive rational number, then $\log_a (mn) = \log_a m + \log_a n$

If $m_1, m_2, m_3, \dots, m_n$ are positive rational numbers, then

$$\log_a (m_1 m_2 m_3 \dots m_n) = \log_a m_1 + \log_a m_2 + \dots + \log_a m_n$$

EXAMPLE 3. The value of $\log_{10} (15)$ is

- a. $\log_{10} 2 - \log_{10} 5$ b. $\log_{10} 5$
 c. $\log_{10} 3 + \log_{10} 5$ d. $\log_{10} 2 + \log_{10} 5$

Sol. c. $\log_{10} 15 = \log_{10} (3 \times 5) = \log_{10} 3 + \log_{10} 5$

(Rule 2) If m and n are positive rational numbers, then

$$\log_a \left(\frac{m}{n} \right) = \log_a m - \log_a n$$

EXAMPLE 4. The value of x in

$$\log x - \log (x - 1) = \log 3 \text{ is}$$

- a. $2/3$ b. $3/2$ c. $1/2$ d. $1/4$

Sol. b. $\log x - \log (x - 1) = \log 3$

$$\Rightarrow \log \frac{x}{x-1} = \log 3 \Rightarrow \frac{x}{x-1} = 3$$

$$\Rightarrow x = 3(x-1) \Rightarrow x = 3x - 3 \Rightarrow -2x = -3$$

$$\therefore x = 3/2$$

(Rule 3) If m and n are positive rational numbers, then,

$$\log_a (m^n) = n \log_a m$$

EXAMPLE 5. The value of x in

$$\log_x 4 + \log_x 16 + \log_x 64 = 12 \text{ is}$$

- a. 1 b. 2 c. 3 d. 4

Sol. b. $\log_x 4 + \log_x 16 + \log_x 64 = 12$

$$\Rightarrow \log_x 2^2 + \log_x 2^4 + \log_x 2^6 = 12$$

$$\Rightarrow 2 \log_x 2 + 4 \log_x 2 + 6 \log_x 2 = 12$$

$$\Rightarrow 12 \log_x 2 = 12 \Rightarrow \log_x 2 = 1$$

$$\therefore x = 2$$

EXAMPLE 6. The value of x in $\frac{\log 256}{\log 16} = \log x$ is

- a. 1 b. 10 c. 100 d. 0

Sol. c. $\frac{\log 256}{\log 16} = \frac{\log 16^2}{\log 16} = \frac{2 \log 16}{\log 16} = 2$

$$\Rightarrow \log x = 2 \Rightarrow \log_{10} x = 2 \quad [\because \log x = \log_{10} x]$$

$$\therefore x = 10^2 = 100$$

(Rule 4) If m is a positive rational number and a, b are positive real numbers i.e. $a \neq 1$ and $b \neq 1$, then

$$\log_a m = \frac{\log_b m}{\log_b a}$$

EXAMPLE 7. The value of

$$\log_3 4 \cdot \log_4 5 \cdot \log_5 6 \cdot \log_6 7 \cdot \log_7 8 \cdot \log_8 9 \text{ is}$$

- a. 1 b. 2 c. -1 d. None of these

Sol. b. $\log_3 4 \cdot \log_4 5 \cdot \log_5 6 \cdot \log_6 7 \cdot \log_7 8 \cdot \log_8 9$

$$= \frac{\log 4}{\log 3} \times \frac{\log 5}{\log 4} \times \frac{\log 6}{\log 5} \times \frac{\log 7}{\log 6} \times \frac{\log 8}{\log 7} \times \frac{\log 9}{\log 8}$$

$$= \frac{\log 9}{\log 3} = \frac{\log 3^2}{\log 3} = \frac{2 \log 3}{\log 3} = 2$$

SOME USEFUL RESULTS

- logarithm of 1 with any base is always zero i.e. $\log_a 1 = 0$
- logarithm of base to itself is always 1 i.e. $\log_a a = 1$
- $x > y \Rightarrow \log_a x > \log_a y$ for $a > 1$
- $x > y \Rightarrow \log_a x < \log_a y$ for $0 < a < 1$
- for $x > 0, 1 \neq a > 0, \log_a n(x) = \frac{1}{n} \log_a x$
- for $1 \neq a, b > 0, x > 0, a \log_b x = x \log_b a$
- If ' a ' is a positive real number and n is a positive rational number, then
 - (i) $\log_a n^x = \frac{x}{y} \log_a n$.
 - (ii) $a^{\log_a n} = n$
 - (iii) $\log_a n = \log_{a^2} n^2 = \log_{a^3} n^3 = \log_{a^m} n^m$

Common Logarithm and Natural Logarithm

Logarithm can have any positive base other than 0. But there are two most important base which are generally used.

1. **Common Logarithm** Logarithm to the base '10' is called common logarithm. It is also called Brigg's logarithm.

$$\text{e.g. } \log 100 = \log_{10} 100 = 2$$

$$\log 1000 = \log_{10} 1000 = 3$$

2. **Natural Logarithm** Logarithm to the base e is called natural logarithm. It is also called Napier logarithm. $\log_e x$ is usually denoted by $\ln x$. e is irrational number between 2 and 3.

$$\ln x = y \Rightarrow e^y = x$$

ANTILOGARITHM

The positive number ' a ' is called the antilogarithm of a number b , if $\log a = b$. If ' a ' is antilogarithm of b , it is written as $a = \text{antilog } b$.

$$\text{So, } a = \text{antilog } b \Leftrightarrow \log a = b$$

Characteristic and Mantissa of a Logarithm

The logarithm of positive real number ' n ' consists of two parts

1. The integral part is known as the characteristic. It is always an integer (positive, negative or zero).
2. The decimal part is called as the mantissa. The mantissa is never negative and is always less than one.

To Find the Characteristic

Case I The characteristic of the log of a number greater than 1 is positive and numerically one less than the number of digits to the left of the decimal part.

e.g.

Value	Characteristic
$\log 3.257$	$1 - 1 = 0$
$\log 32.57$	$2 - 1 = 1$
$\log 325.7$	$3 - 1 = 2$

Case II The characteristic of log of a number less than 1 is negative and numerically one more than the number of zeroes immediately after the decimal point. It is represented by bar over the digit.

The bar over the characteristic indicates that it is negative.

e.g.

Value	Characteristic
$\log 0.3257$	$-(0 + 1) = -1$ i.e. $\bar{1}$
$\log 0.03257$	$-(1 + 1) = -2$ i.e. $\bar{2}$
$\log 0.003257$	$-(2 + 1) = -3$ i.e. $\bar{3}$

Rules for Inserting Decimal Point

Two rules are used for inserting a decimal point.

Rule 5 When the characteristic of the logarithm is positive, we insert the decimal point after the $(n + 1)$ th digit, where n is the characteristic.

Rule 6 When the characteristic of the logarithm is negative, we insert the decimal point such that the

first significant figures is at n th place, where $\bar{n}$ is the characteristic.

EXAMPLE 8. The characteristic of the logarithm of the number 0.00000014 is

- a. 1 b. 7
c. -7 d. None of these

Sol. c. $0.00000014 = 1.4 \times \frac{1}{10000000} = 1.4 \times 10^{-7}$

$\therefore$ The characteristic of $\log 0.00000014 = -7$

EXAMPLE 9. The characteristic of the logarithm of the number 8.188 is

- a. 2 b. 1
c. -1 d. 0

Sol. d. $8.188 = 8.188 \times 1 = 8.188 \times 10^0$

$\therefore$ The characteristic of $\log 8.188 = 0$

EXAMPLE 10. The characteristic of the logarithm of the number 566.37 is

- a. 2 b. -2
c. 4 d. None of these

Sol. a. $566.37 = 56637 \times 10^{-2}$

$\therefore$ The characteristic of $\log 566.37 = 2$

EXAMPLE 11. The characteristic of the logarithm of the number 313 is

- a. 3 b. 2
c. 1 d. 0

Sol. b. $313 = 3.13 \times 10^2$

$\therefore$ The characteristic of $\log 313 = 2$

SOME IMPORTANT POINTS

1. The base of a logarithm is never taken as zero and negative.
2. Log of negative integers are not defined and also $\log_e 0$ is not defined.
3. Logarithmic function is positive as well as negative but exponential function is always positive.